

»» Chapter XIV ««

COMPLEMENTS TO THE CAUCHY THEORY

PREFATORY REMARKS

This chapter is a little Cauchy anthology, the items of which have been chosen to show the broad scope of the theory. We start with Mordell's very ingenious treatment of the celebrated theorem on Gaussian sums (which contains as a by-product the evaluation of the Gauss integral $\int_{-\infty}^{+\infty} \exp(-x^2) dx$). The second to fourth topics are: the problem of characterizing the divisor and the residues of a doubly periodic meromorphic function; the Weierstrass preparation theorem, which sheds light on the local behavior of $\{(z, w): f(z, w) = 0\}$ where f is holomorphic at a point $(a, b) \in \mathbb{C}^2$; the refinement of the Runge theorem permitting us to characterize open subsets O of $\mathbb{C}$ having the property that functions analytic on O may be approximated uniformly in O by polynomials. In the proof to be given of the Weierstrass preparation theorem the notion of the logarithmic derivative will be central.

We consider also two topics important for applications. One is the boundary behavior of integrals of Cauchy type; however we consider the question under severely restricted hypotheses. Our object is not to develop a refined study of the question, but only to indicate its function-theoretic aspect in a simple setting. The other topic is the adaptation of the Cauchy integral theorem to Weierstrass classes. We shall apply the results obtained to an example arising in the theory of ordinary linear differential equations of the second order. The treatment of the Cauchy theorem in this framework is pragmatic in intent and execution.

1. GAUSSIAN SUMS

In this and the following section we shall have occasion to calculate integrals along the *sensed periphery* of a parallelogram. Let us be precise. We recall that given $a, b, c \in \mathbb{C}$, $\text{Im}(\bar{b}c) \neq 0$, the parallelogram determined by (a, b, c) is the image of $[0, 1]^2$ with respect to $\varphi(s, t) = a + bs + ct$. By the integral of f (suitably given) along the (a, b, c) -sensed† periphery of this parallelogram is meant the sum of the integrals of f along

$$\overrightarrow{(a)(a+b)}, \quad \overrightarrow{(a+b)(a+b+c)}, \quad \overrightarrow{(a+b+c)(a+c)}, \quad \overrightarrow{(a+c)(a)}.$$

It will be important to know the order of $\varphi(s, t)$, $0 < s, t < 1$, with respect to the closed curve formed by taking the above four paths in succession and making a periodic extension. It is clear that the order is independent of a . We take $a = 0$. Since the order is unaltered when $b^{-1}\varphi$ replaces φ and $(1, b^{-1}c)$ replaces (b, c) , we are led to the case of the parallelogram determined by $(0, 1, b^{-1}c)$. If $\text{Im}(\bar{b}c) > 0$, the segment $[b^{-1}c, i]$ lies in $\{\text{Im } z > 0\}$ and we see, making a continuous deformation, that the order in question is that associated with the datum $(0, 1, i)$, which is 1. If $\text{Im}(\bar{b}c) < 0$, in the same way we arrive at the order associated with datum $(0, 1, -i)$, that is, -1 .

Mordell's treatment of Gaussian sums. Let n be a positive integer. By the Gaussian sum S_n is meant

$$\sum_0^{n-1} (e^{2\pi i/n})^{k^2}.$$

Gauss gave a simple formula for S_n , many proofs of which are known. Among these the following one of L. J. Mordell [143] is remarkable for its elegance. It contains incidentally—gratis—the evaluation of the Gauss integral. Given $a \in \mathbb{R}$, by $L(a; f)$ we shall understand

$$\lim_{r \rightarrow +\infty} \int_{-r}^r f(a + te^{\pi i/4}) e^{\pi i/4} dt,$$

assuming, of course, f admissible. The interpretation in terms of line integrals is obvious. Mordell's proof starts with the introduction of the function f meromorphic on $\mathbb{C}$ which is simply

$$\sum_0^{n-1} \exp \left[2\pi i \frac{(z+k)^2}{n} \right]$$

† We shall use “sensed” in place of “ (a, b, c) -sensed.”

divided by $e^{2\pi iz} - 1$. An immediate calculation shows that

$$f(z+1) - f(z) = e^{2\pi iz^2/n} (e^{2\pi iz} + 1), \quad (1.1)$$

the standard conventions prevailing. We introduce $I(r)$, the integral of f along the sensed periphery of the parallelogram determined by

$$(-\tfrac{1}{2} - re^{\pi i/4}, 1, 2re^{\pi i/4}),$$

r positive, and verify with the aid of the residue theorem and the observation concerning order made in the first paragraph of this section that $I(r) = S_n$. On noting the existence of $L(\tfrac{1}{2}; f)$ and $L(-\tfrac{1}{2}; f)$, and observing that

$$\lim_{+\infty} \int_{-1/2}^{1/2} f(t - re^{\pi i/4}) dt = \lim_{+\infty} \int_{-1/2}^{1/2} f(t + re^{\pi i/4}) dt = 0,$$

we conclude that

$$S_n = L(\tfrac{1}{2}; f) - L(-\tfrac{1}{2}; f). \quad (1.2)$$

By (1.1) the right side of (1.2) is equal to

$$L\left[-\frac{1}{2}; \exp\left(\frac{2\pi i}{n} z^2\right)\right] + (-i)^n L\left[-\frac{1}{2}; \exp\left[\frac{2\pi i}{n} \left(z + \frac{n}{2}\right)^2\right]\right]. \quad (1.3)$$

We introduce $J(r)$, the integral of $\exp(2\pi iz^2/n)$ along the sensed periphery of the parallelogram determined by $(a - re^{\pi i/4}, b, 2re^{\pi i/4})$, $a \in \mathbf{R}$, b and r positive. By the Cauchy integral theorem $J(r) = 0$. From

$$\lim_{+\infty} J(r) = L\left[a + b; \exp\left(\frac{2\pi iz^2}{n}\right)\right] - L\left[a; \exp\left(\frac{2\pi iz^2}{n}\right)\right]$$

we conclude that $L[a; \exp(2\pi iz^2/n)]$ is independent of a . From this observation applied to (1.3) using (1.2) we obtain

$$\begin{aligned} S_n &= [1 + (-i)^n] L\left[0; \exp\left(\frac{2\pi iz^2}{n}\right)\right] \\ &= [1 + (-i)^n] \frac{1+i}{\sqrt{2}} \sqrt{\frac{n}{2\pi}} \int_{-\infty}^{+\infty} e^{-x^2} dx. \end{aligned} \quad (1.4)$$

By (1.4) and the fact that $S_1 = 1$ we have

$$S_n = \frac{S_n}{S_1} = \frac{1 + (-i)^n}{1 - i} \sqrt{n},$$

and

$$\int_{-\infty}^{+\infty} e^{-x^2} dx = \sqrt{\pi}.$$

2. DOUBLY PERIODIC FUNCTIONS

Let $\gamma, \delta \in \mathbb{C}$ and $\text{Im}(\bar{\gamma}\delta) \neq 0$. Let Ω denote the additive subgroup of $\mathbb{C}$ generated by $\{\gamma, \delta\}$. The object of this section is to penetrate further into the study of the functions f meromorphic on $\mathbb{C}$ which satisfy $f(z + \omega) = f(z)$, $\omega \in \Omega$. Such functions are termed the *meromorphic doubly periodic functions associated with Ω* . Their totality $D(\Omega)$ constitutes a field.

Let $a \in \mathbb{C}$. It is immediate from Ex. 3.6, Ch. X, that $z \in \mathbb{C}$ admits a unique representation of the form

$$a + \gamma s + \delta t + \omega$$

where $0 \leq s, t < 1$, $\omega \in \Omega$. We term the parallelogram determined by (a, γ, δ) a *fundamental parallelogram of Ω* . It follows from the observation just made and the theorem of Liouville that if $f \in D(\Omega)$ is entire, then f is constant.

In the exercises that follow we shall be concerned with the properties of $\partial_f(z)$ and $\text{res}(z; f)$ for $f \in D(\Omega)$ not the constant 0.

EXERCISES

2.1. For such f we have: $\partial_f(z + \omega) = \partial_f(z)$, $\text{res}(z + \omega; f) = \text{res}(z; f)$. Furthermore neither $\{\partial_f(z) \neq 0\}$ nor $\{\text{res}(z; f) \neq 0\}$ clusters at a point of $\mathbb{C}$.

2.2. By choosing a so that $f(z) \neq 0, \infty$ for all z belonging to the frontier of the parallelogram P determined by (a, γ, δ) and integrating $f'(z)$, $f'(z)/f(z)$, $z f'(z)/f(z)$ along the sensed periphery of P conclude that given any $a \in \mathbb{C}$ and $\{\alpha, \beta\}$ generating Ω , with Q denoting

$$\{a + \alpha s + \beta t: 0 \leq s, t < 1\},$$

we have:

$$(a) \sum_{z \in Q} \text{res}(z; f) = 0,$$

$$(b) \sum_{z \in Q} \partial_f(z) = 0,$$

$$(c) \sum_{z \in Q} z \partial_f(z) \in \Omega.$$

Apply (b) to $f - c$ where f is not constant and c is a complex constant to conclude that $\sum_{f|Q(z)=k} n(z; f)$ is independent of $k \in \mathbb{C}$.

2.3. Show that condition (a) and the results of Ex. 2.1 permit us to characterize the distribution of the residues of $f \in D(\Omega)$. More precisely, let $\rho: \mathbb{C} \rightarrow \mathbb{C}$ satisfy (1) $\{\rho(z) \neq 0\}$ clusters at no point of $\mathbb{C}$, (2) $\rho(z + \omega) = \rho(z)$, $\omega \in \Omega$, (3) $\sum_{z \in Q} \rho(z) = 0$. Then there exists $f \in D(\Omega)$ such that $\text{res}(z; f) = \rho(z)$, $z \in \mathbb{C}$. Hint: Let ζ denote the Weierstrass zeta function associated with Ω (cf. Ex. 6.10, Ch. VIII) and consider

$$\sum_{b \in Q} \rho(b) \zeta(z - b).$$

Show with the aid of the Weierstrass ζ -function associated with Ω and its successive derivatives that there exists $f \in D(\Omega)$ with assigned principal parts $P_b[(z - b)^{-1}]$

where P_b is a polynomial, $b \in Q$, subject only to the conditions that P_b is not the zero constant for only a finite set of b and that

$$\sum_{b \in Q} \text{res}\{b; P_b[(z-b)^{-1}]\} = 0.$$

2.4. Show that if $\partial: C \rightarrow Z$ satisfies (1) $\{\partial(z) \neq 0\}$ clusters at no point of C , (2) $\partial(z + \omega) = \partial(z)$, $\omega \in \Omega$, (3) $\sum_{z \in Q} \partial(z) = 0$, (4) $\sum_{z \in Q} z \partial(z) \in \Omega$, then there exists $f \in D(\Omega)$ satisfying $\partial_f = \partial$. Further such f are determined up to a nonzero multiplicative constant. Hint: With σ the Weierstrass sigma function associated with Ω , $c \in Q$, $d = \sum_{z \in Q} z \partial(z)$, consider

$$f(z) = [\sigma(z-c)]^{\theta(c)-1} \sigma(z-c+d) \prod_{b \in Q-(c)} [\sigma(z-b)]^{\theta(b)}$$

and verify that f so specified has the stated properties.

2.5. Let p denote the Weierstrass p -function associated with Ω . Suppose that a is taken as 0 in the definition of Q . Show that the zeros of p' in Q are precisely $\alpha/2$, $\beta/2$, $(\alpha + \beta)/2$ and that each has multiplicity 1. Hint: Use the fact that p is even.

2.6. Suppose that A and B are positive and that $\alpha = A$, $\beta = iB$. Show that p takes real values on $\text{fr}[0, A/2; 0, B/2] - \{0\}$ and that p maps $\text{int}[0, A/2; 0, B/2]$ univalently onto $\{\text{Im } z < 0\}$.

3. THE WEIERSTRASS PREPARATION THEOREM

This theorem (also known by its German name “Vorbereitungssatz”) gives a useful insight into the local behavior of the sets of constancy of a holomorphic function of several complex variables. We treat here the case of two complex variables.† We remark that the corresponding situation in the real C^∞ theory is vastly more complicated. In the exposition that follows we assume that f is holomorphic on $[\Delta(0; r)]^2$ and not constant and that $f(0, 0) = 0$. It is clear that the local study of a function F holomorphic at (a, b) may be reduced to the normalized case by considering the translated function $F(a + z, b + w) - F(a, b)$. The Weierstrass preparation theorem asserts

Theorem 3.1: *There exists ρ , $0 < \rho < r$, such that*

$$f(z, w) = z^m \left[w^n + \sum_{k=0}^{n-1} a_k(z) w^k \right] g(z, w), \quad (3.1)$$

$(z, w) \in [\Delta(0; \rho)]^2$, where (1) $m, n \in \mathbb{N}$, $m + n > 0$, (2) a_k is analytic in $\Delta(0; \rho)$ and $a_k(0) = 0$ for $k = 0, \dots, n-1$, and (3) g is holomorphic in $[\Delta(0; \rho)]^2$ and does not take the value 0.

† For the general situation, cf. [18].

We see that the preparation theorem reduces the study of the set $\{f(z, w) = 0\}$ to the study of the zero sets of the first two factors and for these we have a fair amount of information.†

PROOF: We start with the representation

$$f(z, w) = \sum_0^{\infty} b_k(w) z^k,$$

b_k analytic on $\Delta(0; s)$. Let m denote the minimum of the k for which b_k is not the constant 0. The proof reduces to the case $m = 0$ and we continue under this assumption. If $b_0(0) \neq 0$, the theorem follows at once. Suppose that $b_0(0) = 0$ and let n denote the multiplicity of b_0 at 0. Let s , $0 < s < r$, be such that $b_0(w) \neq 0$, $0 < |w| \leq s$. Now take t , $0 < t < r$, such that $f(z, w) \neq 0$ for $|z| \leq t$, $|w| = s$. For each allowed z the equation $f(z, w) = 0$ has n roots in $\Delta(0; s)$ counted by multiplicity. For each such z let

$$\{\alpha_k(z): 1 \leq k \leq n\} = \{w: f(z, w) = 0, |w| < s\}$$

where the number of k such that $\alpha_k(z)$ is a given member ω of the set on the right is precisely the multiplicity at ω of the analytic function assigning to w the value $f(z, w)$.

We proceed by defining $P(z, w)$ as

$$\prod_1^n [w - \alpha_k(z)] = w^n + \sum_0^{n-1} a_k(z) w^k,$$

$|z| < t$. For each such z , the function assigning to w the value

$$\frac{f_w(z, w)}{f(z, w)} - \frac{P_w(z, w)}{P(z, w)} \quad (3.2)$$

is analytic at each point of $\overline{\Delta(0; s)} - \{\alpha_1(z), \dots, \alpha_n(z)\}$ and has removable singularities at the $\alpha_k(z)$. When $w \in \Delta(0; s) - \{\alpha_1(z), \dots, \alpha_n(z)\}$, the value given in (3.2) is equal to

$$\frac{1}{2\pi i} \oint_{C(0; s)} \frac{f_w(z, \omega)}{f(z, \omega)} \frac{1}{\omega - w} d\omega. \ddagger \quad (3.3)$$

The function defined by (3.3) on $\Delta(0; t) \times \Delta(0; s)$ is holomorphic on this set. Now there exists q , $0 < q < s$, such that $f(z, w) \neq 0$ for $|z| < t$, $q < |w| < s$.

† For the zero set of the second factor, cf. §3, Ch. XVIII.

‡ Cf. Theorem 1.2, Ch. VII.

Hence by (3.2) and the integral representation (3.3) we see that $P_w(z, w)/P(z, w)$ is holomorphic in $\Delta(0; t) \times \{q < |w| < s\}$. We have for $|z| < t$ the representation

$$\frac{P_w(z, w)}{P(z, w)} = \sum_1^{\infty} c_k(z) w^{-k}, \quad q < |w| < +\infty,$$

where each c_k is analytic on $\Delta(0; t)$, as we see using the observation of the preceding sentence and the formula

$$c_k(z) = \frac{1}{2\pi i} \oint_{C(0; \lambda)} \frac{P_w(z, w)}{P(z, w)} w^{k-1} dw,$$

$q < \lambda < s$. From the identity

$$P_w(z, w) = P(z, w) \sum_1^{\infty} c_k(z) w^{-k},$$

$|z| < t, q < |w| < +\infty$, we see on multiplying out and equating corresponding coefficients that the coefficients $a_0, \dots, a_{n-1}$ entering in $P(z, w)$ above are analytic on $\Delta(0; t)$. For each $z \in \Delta(0; t)$ we have a representation of the form

$$f(z, w) = P(z, w) g(z, w), \quad |w| < r,$$

where the function assigning $g(z, w)$ to w is analytic on $\Delta(0; r)$. Applying the Cauchy integral formula for a disk to g we have

$$g(z, w) = \frac{1}{2\pi i} \oint_{C(0; s)} \frac{f(z, \omega)}{P(z, \omega)} \frac{1}{\omega - w} d\omega,$$

$|z| < t, |w| < s$, so that g is holomorphic in $[\Delta(0; \rho)]^2$ where $\rho = \min\{s, t\}$. Further $g(z, w) \neq 0$ for $(z, w) \in [\Delta(0; \rho)]^2$. We note that for $z = 0$ we have $\alpha_k(0) = 0, k = 1, \dots, n$, and hence $a_k(0) = 0, k = 0, \dots, n-1$. The preparation theorem is seen to follow.

EXERCISES

3.1. Treat the uniqueness of the representation (3.1).

3.2. Extend the preparation theorem to the case of a function f of $n > 2$ complex variables which is holomorphic at $(0, \dots, 0) \in \mathbb{C}^n$ and satisfies $f(0, \dots, 0) = 0$ and

$$f(0, \dots, 0, z_n) = z_n^m g(z_n),$$

z_n small, m a positive integer, g analytic at 0, $g(0) \neq 0$.

3.3. Construct an example of a C^∞ real-valued function of two real variables

which has the property that some point of the zero set belongs to the frontier of the interior of the zero set.

4. RUNGE'S THEOREM ON POLYNOMIAL APPROXIMATION

We return to the Runge theorem concerning approximation by rational functions (Theorem 8.1, Ch. V) and apply it to obtain the following theorem of Runge on the possibility of uniform polynomial approximation.

Theorem 4.1: *Let O be an open subset of $\mathbb{C}$. Then the connectedness of $\hat{\mathbb{C}} - O$ is a necessary and sufficient condition for O to have the property that given f analytic on O , K compact $\subset O$, and ε a positive number, there exists a polynomial p such that*

$$|f(z) - p(z)| < \varepsilon, \quad z \in K.$$

PROOF: Let us first consider the necessity of the condition. Suppose contrary to the assertion that $\hat{\mathbb{C}} - O$ is not connected. Then there exist closed subsets A and B of $\hat{\mathbb{C}}$, which are nonempty, disjoint, and have union $\hat{\mathbb{C}} - O$. We suppose that $\infty \in B$. If $B = \{\infty\}$, and a positive number r is so chosen that $A \subset \Delta(0; r)$, then $(z - a)^{-1}$, where a is a given point of A , may be approximated uniformly on $C(0; r)$ by a polynomial thanks to the hypothesis of the necessity part of the theorem. Introducing a sequence of polynomials tending uniformly to $(z - a)^{-1}$ on $C(0; r)$, we see that it is uniformly Cauchy on $C(0; r)$ and hence by the boundary maximum principle for analytic functions it is uniformly Cauchy on $\overline{\Delta(0; r)}$. The sequence tends uniformly on $\overline{\Delta(0; r)}$ to a function f continuous on $\overline{\Delta(0; r)}$ and analytic in $\Delta(0; r)$. Further $f(z) = (z - a)^{-1}$, $|z| = r$. Hence $f(z)(z - a) - 1 = 0$, $|z| = r$, and consequently $f(z)(z - a) = 1$, $|z| \leq r$, and this is obviously false as we see on setting $z = a$.

The argument just given holds with slight modification when B contains points of $\mathbb{C}$. In this case we introduce $m = \text{dist}(A, B - \{\infty\})$, which is a positive number, and take n a positive integer so that $\sqrt{2} < mn$. We let S denote the union of those squares $[n^{-1}j, n^{-1}(j+1)]$; $[n^{-1}k, n^{-1}(k+1)]$, $j, k \in \mathbb{Z}$, which have points in common with A . Now S is a compact subset of $\mathbb{C}$, $S \cap B = \emptyset$, and $\text{fr } S \subset O$. We now introduce $(z - a)^{-1}$ where a is a given point of A , and we repeat the above argument with $\text{fr } S$ replacing $C(0; r)$.

The asserted necessity of the connectedness of $\hat{\mathbb{C}} - O$ follows.

Sufficiency. We base the proof on the Runge theorem concerning approximation by rational functions (Theorem 8.1, Ch. V) and the following lemma:

Let K be a given compact subset of $\mathbb{C}$. Let E denote the set of $a \in \mathbb{C} - K$ such that $(z - a)^{-1}$ may be approximated uniformly on K by a polynomial. Then E is the unbounded component of $\mathbb{C} - K$.

To be exact, when we say that $(z - a)^{-1}$ may be approximated uniformly on K by a polynomial, we mean that for each positive ε there exists a polynomial p such that

$$|(z - a)^{-1} - p(z)| < \varepsilon, \quad z \in K.$$

This language will be used generally without further gloss.

PROOF OF THE LEMMA: We put aside the trivial case where K is empty. If $a \in \mathbb{C} - K$ and $|b - a| < \text{dist}(\{a\}, K)$, then using the Laurent expansion of $(z - b)^{-1}$ in the annulus $\{|b - a| < |z - a| < +\infty\}$ we see that $(z - b)^{-1}$ may be approximated uniformly on K by a polynomial in $(z - a)^{-1}$. If $a \in E$ and P is a polynomial, then $P[(z - a)^{-1}]$ may be approximated uniformly on K by a polynomial as we see on noting that $(z - a)^{-k}$, $k \in \mathbb{N}$, shares with $(z - a)^{-1}$ the property that it may be approximated uniformly on K by a polynomial. Hence if $a \in E$ and $|b - a| < \text{dist}(\{a\}, K)$, then $b \in E$. Consequently, E is open.

Let us now show that E is closed in the sense of the relative topology of $\mathbb{C} - K$. Indeed, if $b \in (\mathbb{C} - K) \cap \bar{E}$, there exists $a \in E$ such that $2|b - a| < \text{dist}(\{b\}, K)$. We have

$$|b - a| < \text{dist}(\{b\}, K) - |b - a| \leq \text{dist}(\{a\}, K).$$

Hence $(z - b)^{-1}$ may be approximated uniformly on K by a polynomial. The asserted property of E follows.

Clearly all points of $\mathbb{C}$ with large modulus belong to E . Hence E contains the unbounded component of $\mathbb{C} - K$.

Suppose that $a \in \Omega$, a bounded component of $\mathbb{C} - K$, and that $(z - a)^{-1}$ admitted uniform approximation on K by a polynomial. Repeating the argument used in treating the first half of the present theorem we infer the existence of f continuous on $\bar{\Omega}$ and analytic in Ω satisfying $(z - a)f(z) = 1$, $z \in \text{fr } \Omega$. The same contradiction is reached. Hence $\Omega \cap E = \emptyset$. We conclude that E is the unbounded component of $\mathbb{C} - K$.

The sufficiency proof is now readily completed on appeal to the earlier Runge theorem (Theorem 8.1, Ch. V). It suffices to note that $\hat{\mathbb{C}} - O \subset E \cup \{\infty\}$, where now E is taken relative to a given compact subset K of O , and that the finite poles of an approximant furnished by Theorem 8.1, Ch. V, lie in E . The remainder of the argument is routine.

EXERCISE

4.1. Suppose that $f(z) = 0$, $z \in \Delta(-1; 1)$, and $f(z) = 1$, $z \in \Delta(1; 1)$. Show that Theorem 4.1 is applicable.

5. INTEGRALS OF CAUCHY TYPE, CAUCHY PRINCIPAL VALUE

Let $-\infty < a < b < +\infty$ and suppose that g is a complex-valued member of $C'[a, b]$. We are concerned with the limiting behavior of the integral of Cauchy type

$$f(z) = \int_a^b \frac{g(t)}{t-z} dt$$

at points of $]a, b[$. Suppose that $c \in]a, b[$. We consider

$$\lim_0 \left[\int_a^{c-\varepsilon} \frac{g(t)}{t-c} dt + \int_{c+\varepsilon}^b \frac{g(t)}{t-c} dt \right], \quad (5.1)$$

$0 < \varepsilon < \min\{c-a, b-c\}$. Setting $g(t) = [g(t) - g(c)] + g(c)$, we see that the limit in question is

$$\int_a^b h(t) dt + g(c) \log \frac{b-c}{c-a}, \quad (5.2)$$

where $h(t) = [g(t) - g(c)]/(t-c)$ when $t \neq c$ and $h(c) = g'(c)$. We denote the limit by $P(c; g)$. We are concerned here with the notion of a *Cauchy principal value* of an improper integral. Generally, if F is a continuous map of $[a, b] - \{c\}$ into $\mathbb{C}$ —the continuity condition may be replaced by a weaker one—then by the *Cauchy principal value* of

$$\int_a^b F(x) dx$$

is meant

$$\lim_0 \left[\int_a^{c-\varepsilon} F dx + \int_{c+\varepsilon}^b F dx \right],$$

assumed existent.

Suppose that $g = G| [a, b]$ where G is analytic at each point of $[a, b]$. We shall show that

$$\lim_c f| \{ \text{Im } z > 0 \} = P(c; g) + \pi i g(c). \quad (5.3)$$

Indeed, for ε small and positive we have by the Cauchy integral theorem

$$f(z) = \int_a^{c-\varepsilon} \frac{g(t)}{t-z} dt + \int_{-\pi}^0 \frac{G(c + \varepsilon e^{i\theta})}{c + \varepsilon e^{i\theta} - z} i\varepsilon e^{i\theta} d\theta + \int_{c+\varepsilon}^b \frac{g(t)}{t-z} dt$$

and we conclude that the desired limit is

$$\int_a^{c-\varepsilon} \frac{g(t)}{t-c} dt + \int_{c+\varepsilon}^b \frac{g(t)}{t-c} dt + \int_{-\pi}^0 G(c + \varepsilon e^{i\theta}) i d\theta$$

and hence that (5.3) holds.

Formula (5.3) is also valid when g satisfies the original C' hypothesis. For by the Weierstrass (polynomial) approximation theorem we may approximate g' uniformly on $[a, b]$ by a polynomial. If p is a polynomial satisfying $p(a) = g(a)$, we have

$$|P(c; g) - P(c; p)| \leq M(b-a) \left(1 + \log \frac{b-c}{c-a}\right), \quad (5.4)$$

where $M = \max\{|g'(x) - p'(x)|: a \leq x \leq b\}$, as we see by reference to (5.2) and a routine estimate of $P(c; g)$ in the case where $g(a) = 0$. Further setting $q(t) = g(t) - p(t)$, we obtain from

$$\int_a^b \frac{q(t)}{t-z} dt = \int_a^b \frac{q(t) - q(x)}{t-x} \frac{t-x}{t-z} dt + q(x) \int_a^b \frac{1}{t-z} dt$$

the inequality

$$\left| \int_a^b \frac{q(t)}{t-z} dt \right| \leq M(b-a) \left(1 + \log \left| \frac{b-z}{a-z} \right| + \pi\right) \quad (5.5)$$

when $\operatorname{Im} z \neq 0$, $a \leq x = \operatorname{Re} z \leq b$. Using (5.4) and (5.5) we conclude that

$$\limsup_c |f(z) - P(c; g) - \pi i g(c)| \leq 2M(b-a) \left(1 + \pi + \log \frac{b-c}{c-a}\right),$$

and hence that (5.3) holds under the C' hypothesis since p may be so chosen that M is arbitrarily small. It is to be observed that a polynomial approximant yielded an analytic approximant—a fact that put the Cauchy apparatus at our disposal.

A corresponding formula for the case of the lower half-plane holds. It is

$$\lim_c f | \{\operatorname{Im} z < 0\} = P(c; g) - \pi i g(c).$$

The above developments extend to vector-valued analytic functions.

EXERCISES

5.1. Given a, b as above and h positive. Show with the aid of the Cauchy integral formula for the closure of

$$[a_2, b_2; c_2, d_2] - [a_1, b_1; c_1, d_1],$$

where $a_2 < a_1 < b_1 < b_2$ and $c_2 < c_1 < d_1 < d_2$, that if f is analytic on

$$A(h) = \text{int}[a - h, b + h; -h, h] - [a, b],$$

then there exists a unique (f_i, f_e) , where f_i is analytic on $\text{int}[a - h, b + h; -h, h]$ and f_e is analytic on $\hat{C} - [a, b]$ and satisfies $f_e(\infty) = 0$, such that

$$f(z) = f_i(z) + f_e(z), \quad z \in A(h).$$

5.2. Suppose that $g = G|_{[a, b]}$ where G is analytic at each point of $[a, b]$. Let

$$H(z, w) = \frac{G(z) - G(w)}{z - w}, \quad z \neq w,$$

and

$$H(z, z) = G'(z), \quad z, w \in A(h), \quad h \text{ small.}$$

Then $\int_a^b H(z, t) dt$ is analytic on $A(h)$. Further

$$\int_a^b H(z, t) dt = \frac{1}{2\pi i} \int_{\beta(h)} \frac{G(w) \lambda(w)}{w - z} dw, \quad z \in A(h),$$

where $\lambda(w) = \text{Log}[(w - a)/(w - b)]$ and $\beta(h)$ is the positively sensed periphery of $[a - h, b + h; -h, h]$. Conclude that if $\int_a^b H(x, t) dt = 0$, $a < x < b$, then g is constant. Show that if φ is analytic on $A(h)$, then with $G = (\varphi/\lambda)_i$ we have

$$\int_a^b H(z, t) dt = \varphi(z), \quad z \in A(h).$$

[Carleman, cf. [24, p. 142].] Relate this problem with the subject of Cauchy principal value.

6. THE CAUCHY INTEGRAL THEOREM FOR WEIERSTRASS CLASSES

Given a Weierstrass class $\mathbf{W}$ we introduce two complex-valued functions having domain $\mathbf{W}$: One is the *center function* c assigning a to $e = (a, P) \in \mathbf{W}$; the other is the *value function* v assigning $P(a)$ to e . The terminology is that of Saks and Zygmund (cf. [108]). Let us now consider $\varphi: [0, 1]^2 \rightarrow \mathbf{W}$ having the following properties: (1) $c \circ \varphi$ is continuous on $[0, 1]^2$, (2) $\varphi(s, t)$ is a direct continuation of $\varphi(s_0, t_0)$ for (s, t) sufficiently near (s_0, t_0) , (s_0, t_0) arbitrary in $[0, 1]^2$. It is easily concluded that there exists δ positive such that $\varphi(s_1, t_1)$ and $\varphi(s_2, t_2)$ are direct continuations of one another whenever the distance between (s_1, t_1) and (s_2, t_2) is less than δ .

Suppose now that $c \circ \varphi$ taken along the edges of $[0, 1]^2$ defines a rectifiable

path. More explicitly, we define C on $[0, 4]$ by

$$\begin{aligned} C(t) &= c \circ \varphi(t, 0), & 0 \leq t \leq 1, \\ &= c \circ \varphi(1, t-1), & 1 \leq t \leq 2, \\ &= c \circ \varphi(3-t, 1), & 2 \leq t \leq 3, \\ &= c \circ \varphi(0, 4-t), & 3 \leq t \leq 4, \end{aligned}$$

and suppose that C is rectifiable. We define V having domain $[0, 4]$ similarly, v replacing c in the definition. We have a direct extension of the Cauchy theorem for a singular cell, namely, that under the stated assumptions

$$\int_0^4 V \, dC = 0.$$

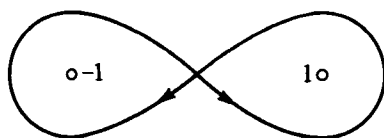
The proof is a direct paraphrase of that of Theorem 3.1, Ch. V. We take n a positive integer such that $\sqrt{2} < n\delta$, reference being made to a positive number δ allowed in the sense of the first paragraph of this section, and subdivide $[0, 1]^2$ as earlier into n^2 squares. Here we introduce in place of g_{jk} a primitive of the second component of $\varphi(j/n, k/n)$, $j, k = 0, \dots, n-1$. The details then go as in the earlier proof.

A similar result holds when φ satisfies $\varphi(0, t) = \varphi(1, t)$, $0 \leq t \leq 1$, and $C_1(t) = c \circ \varphi(t, 0)$, $C_2(t) = c \circ \varphi(t, 1)$, $0 \leq t \leq 1$, are rectifiable. Setting $V_1(t) = v \circ \varphi(t, 0)$ and $V_2(t) = v \circ \varphi(t, 1)$ we have

$$\int_0^1 V_1 \, dC_1 = \int_0^1 V_2 \, dC_2. \quad (6.1)$$

It is well to consider an example. The one below arises in the theory of linear differential equations of the second order. Essentially our task is one of interpretation.

EXAMPLE: The integral



$$\int_{\Gamma} e^{zw}(w^2 - 1)^{\alpha} dw,$$

FIG.14.1

where $\alpha \in \mathbb{C} - \mathbb{N}$ and Γ is a rectifiable closed curve in $\mathbb{C} - \{-1, 1\}$ satisfying the order condition $O(-1; \Gamma) = -1$, $O(1; \Gamma) = 1$, as indicated in the figure. Since we are concerned with continuous determinations of the integrand, or

more strictly of $\exp[z \Gamma(t)]$ times an α th power of $[\Gamma(t)]^2 - 1$, as is intended, there is no difficulty in definition up to a constant factor of the form $e^{2\pi iak}$, $k \in \mathbf{N}$. For the present consideration the important point is the use of the Cauchy theory to establish independence of path results.

Specifically, suppose that Γ is deformed into a rectifiable closed curve Γ_1 by ψ , a continuous map of $\mathbf{R} \times [0, 1]$ into $\mathbf{C} - \{-1, 1\}$, satisfying

$$\begin{aligned}\psi(s+1, t) &= \psi(s, t), & (s, t) \in \mathbf{R} \times [0, 1]; \\ \psi(s, 0) &= \Gamma(s), & \psi(s, 1) = \Gamma_1(s), & s \in \mathbf{R}.\end{aligned}$$

There exists a continuous logarithm l of $\psi^2 - 1$. We have $l(s+1, t) = l(s, t)$, $(s, t) \in \mathbf{R} \times [0, 1]$. Each continuous map f of $\mathbf{R} \times [0, 1] \times \mathbf{C}$ into $\mathbf{C}$ satisfying

$$f(s, t, z) = \exp[z \psi(s, t)] g(s, t)$$

where $g(s, t)$ is an α th power of $[\psi(s, t)]^2 - 1$ is of the form

$$e^{2\pi iak} \exp[z \psi(s, t) + \alpha l(s, t)], \quad (6.2)$$

where k is an integer independent of admitted (s, t, z) . We observe that for fixed z there exists φ (unique) mapping $[0, 1]^2$ into a Weierstrass class which meets the conditions (1) and (2) of the first paragraph of this section as well as (3) $c \circ \varphi(s, t) = \psi(s, t)$, $(s, t) \in [0, 1]^2$, and (4) the second component of $\varphi(s, t)$ takes the value $\exp[zw + \alpha \lambda(w)]$ at each point w of some neighborhood of $\psi(s, t)$. Here λ is an analytic logarithm of $w^2 - 1$ with domain a small disk centered at $\psi(s, t)$ which takes the value $l(s, t)$ at $\psi(s, t)$. The verification is routine. We see that

$$v \circ \varphi(s, t) = \exp[z \psi(s, t) + \alpha l(s, t)].$$

By (6.1) we conclude that the integral we are considering is unaltered when Γ is replaced by Γ_1 , where it is understood that use is made of a fixed continuous determination (6.2) to specify the integrals involved. In our present example the integral yields an entire function as we see by fixing $t = 0$.

EXERCISES

6.1. Suppose that $\Gamma(t) = 1 + e^{\pi i(4t-1)}$, $0 \leq t \leq \frac{1}{2}$, $= -1 + e^{\pi i(2-4t)}$, $\frac{1}{2} < t \leq 1$, and that Γ is defined elsewhere on $\mathbf{R}$ so as to be periodic with period 1. Show that if $\operatorname{Re} \alpha > -1$ the functions defined by the integral of the example using a "continuous determination of the integrand along Γ " (that is, $f(s, 0, z)$ of the text) admit a representation of the form

$$(-2ie^{2k\pi i\alpha} \sin \pi\alpha) \int_{-1}^1 \exp[zt + \alpha \log(1-t^2)] dt, \quad k \in \mathbf{Z}.$$

6.2. Let $\alpha \in \mathbb{C}$. Let $\{\omega_1(z), \omega_2(z)\}$ be the pair of roots of $w^2 - 2zw + 1 = 0$ where z lies in an open disk D not containing either of the points $1, -1$. We suppose that ω_1 and ω_2 are analytic. Let Γ denote a rectifiable closed curve lying in $\mathbb{C} - \{0, \omega_1(D), \omega_2(D)\}$ satisfying

$$\begin{aligned} O(0; \Gamma) &= O[\omega_1(z); \Gamma] = 0, \\ O[\omega_2(z); \Gamma] &= -2, \end{aligned}$$

as, for example, in the figure. Interpret the formally given

$$\int_{\Gamma} w^{\alpha} (w^2 - 2zw + 1)^{-1/2} dw.$$

Treat the questions of continuous determination of the integrand, independence of path, analyticity of the function defined by the integral. Investigate the nonvanishing of the integral when z is small and Γ is specified by taking the paths $\gamma_1, \gamma_2, \gamma_3, \gamma_4$ in succession as indicated in the figure and extending periodically.

6.3. Treat $\int_{\Gamma} [(z^2 - 1)(z - 3)]^{1/2} dz$ where (a) Γ is the closed curve of Ex. 6.1, (b) $\Gamma(t) = 2e^{2\pi it}$. In each case reduce the calculation of the integral to that of the integral of a real-valued continuous function with domain $[-1, 1]$.

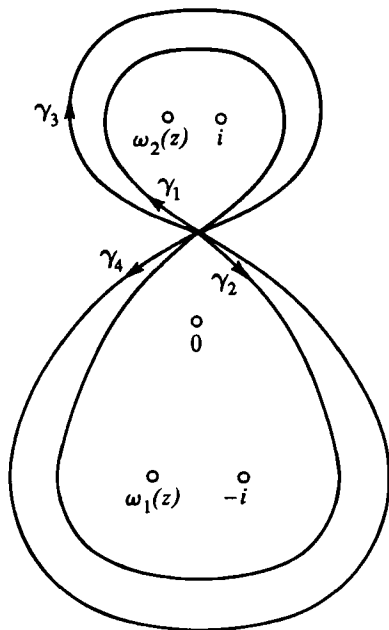


FIG.14.2

There is a wealth of material on integrals of the type considered in this section. For their further study the following references are useful: [13], [34], [45], [97], [130].